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(1) Research Program/Project/Study Title: Application of a Clay Pot Cooling System in Indoor Hydroponics	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Pontevedra
Province:	Capiz
Region:	IV

(5) Classification Research: Development: ☐ Basic ☒ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
(7) Sector/Commodity High Value Crop	(8) Discipline/Field of Specialization Agricultural and Biosystems Engineer

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Hydroponics is a soilless farming technology that utilizes nutrient solution for crop production. The advancement of hydroponic farming technology is mainly influenced by health concerns, environmental factors, water resources and soil-based problems, as well as the desire to improve yield and quality of agricultural crops (Kumari et al., 2018). Apparently today, existing farming systems face innumerable challenges to complement with food sufficiency and sustainability, thereby entailing an increasing demand for hydroponic greenhouse farming (Khan et al., 2018). Sharma et al. (2018) supported by stating that hydroponic growing method is presently gaining global popularity because the method effectively utilizes available resources resulting to quality produce.

Hydroponic system is one of the solutions to provide sufficient, safe and clean food but the challenge is the temperature rise resulting to heat stress for plants. Heat stress is a common problem that must be addressed properly or else death may occur. Studies show that temperature has significant impact in the production of agricultural crops. One of the characteristics of hydroponic cultivation is its ability to control the temperature of the nutrient solution around the root system using heaters or cooling spirals, to increase or decrease the temperature, respectively. The cost-effective modification of root-zone temperature (RZT) as compared to the cooling of the whole greenhouse could be a necessary solution to temperature stress for the crop plants in hydroponics (Al-Rawahy, 2018).

Philippines is a tropical country with the mean annual temperature of 26°C, with warmest month occurs in May with a mean temperature of 28.3°C (PAGASA DOST 2023). Because of the effects of climate change, some part of the region is experiencing drought with high temperature even rainy season. High temperature really affects the production of agricultural crops even inside the greenhouse. The use of ground heat exchanger, air-conditioning unit and water chiller which uses high power consumption of electricity is an additional cost to hydroponics system. Some hydroponic farms in the Philippines preferred to use in-ground cooling of the reservoir due to free energy. However, there is no enough research available to prove the efficiency of in-ground cooling and quality of produce in hydroponics. With the foregoing reasons, the idea of using a clay pot cooling system for hydroponics was generated. The temperature of the nutrient solution could be managed up to the optimum condition favorable for plant growth.

This technology will help not only to increase the yield of the plant but also to save electricity. The beneficial effect of this cooling system to hydroponics will promote and encourage Filipinos to adopt the hydroponic cultivation which is economical and environment-friendly. Moreover, this technology will promote a sustainable production all year round.

Since this technology is not yet applied in the Philippines for hydroponic production, research is needed in order to prove the advantage of this technology. Hence this study will be conducted.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The study generally aims to design, fabricate and evaluate a clay pot cooling system for indoor hydroponics production. Specifically, it aims to:

1. design a clay pot cooling system;
2. fabricate a clay pot cooling system;
3. evaluate a clay pot cooling system in terms of nutrient solution temperature;
4. determine the performance of the system in terms of yield of lettuce (g); and
5. determine the cost of production of using the system.

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(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of -the-art of current technology / information from which the project proposal will take-off).

Evaporative Cooler

Ndukwu. M. C. (2011) stated that an evaporative cooler was developed with clay and other locally available materials. The performance of cooler was evaluated in terms of temperature drop, evaporative effectiveness and cooling capacity. The result showed that the evaporative cooler can reduce the daily maximum ambient temperature from 32 – 40 °C to 24 – 29 °C i.e. a temperature reduction of up to 10 °C and increase the relative humidity of incoming air from 40.3% to 92% for the storage chamber. The cooling efficiency ranged from 20 -92 % with a maximum cooling capacity of 1207W. The cooling capacity and effectiveness of the cooler was higher during the day between the period of 12 -16 hours local time when harvesting and selling of farm produce is done. The evaporative cooler was able to preserve freshly harvested tomato for 19 days before visible color changes and mold spotting appeared and the weight drastically reduced.

In addition, Prabodh Sai Dutt et al. (2016) said Conventional Refrigerators has its own share of problems pertaining to environmental safety, usage and maintenance in rural parts of developing countries. Refrigeration by evaporative cooling has proved to be a successful alternative in providing inexpensive and environment friendly way to preserve food and medicines. Though the concept of evaporative cooling has been in existence from a long time, application of the concept towards preservation of food, agricultural produce and medicines has seen a surge only from past few years.

Prior Art Search

CN103598079A. Nutrient solution cooling system and summer cooling method thereof. The invention discloses a nutrient solution cooling system and a summer cooling method thereof, and belongs to the technical field of soilless cultivation of facility horticulture. According to the cooling system, by the aid of a wet curtain and a fan cooling system, high-temperature nutrient solution exchanges heat with the wet curtain and low-temperature circulating water in a water storage tank of the fan cooling system, the temperature of the nutrient solution is reduced, and the cooled nutrient solution is delivered to a plant root system through a drip irrigation system or a nutrient solution circulating system, so that the temperature of rhizosphere is reduced. In daytime, the nutrient solution exchanges heat by the aid of a wet curtain water circulating system to reduce the temperature, and the cooled nutrient solution is delivered to the plant root system through a sealed cultivation system, so that the rhizospheric environment temperature of a plant is reduced. At night, the rhizospheric environment temperature of the plant is not manually regulated and naturally changes along with temperature drop. The system and the method are simple, low in cost, practical, easy to implement and fine in cooling effect.

KR100407075B1. Nutrient solution bed cooling and heating system using heat pump It maintains the temperature at which crops for nutrient solution cultivation can grow under optimal conditions in summer or winter. It consists of a compressor, a refrigerant pipe through which the refrigerant discharged from the compressor moves, and a water pipe through which water moves by accommodating the refrigerant pipe. A condenser that exchanges heat between the refrigerant and water, an evaporator that exchanges heat with air and the refrigerant that has exchanged heat with water moving from the condenser, and a refrigerant pipe of the compressor are installed to communicate with the refrigerant pipe of the condenser and the evaporator, switching When connected, a four-way valve for switching between cooling and heating that selects cooling or heating by controlling the flow direction of the refrigerant, and water for storing hot and cold water that stores hot or cold water generated when the refrigerant is condensed or evaporated by being communicated with the outlet side of the water transfer pipe of the condenser. A tank and a water pump for supplying hot or cold water that circulates hot or cold water stored in the water tank through cold and hot water pipes under the nutrient solution bed to maintain the temperature of the medium in the nutrient solution bed at a set temperature. Annual production of crops for hydroponic cultivation is possible, which contributes to the increase in farm household income by increasing production volume, and when operating a heat pump system that uses electricity as driving energy, hot water or cold water with a calorific value equivalent to 2 to 3 times compared to the amount of electricity used.

KR20190010842A. A Control System of Temperature for Hydroponic Crops using Peltier Module. The present invention relates to a temperature preservation device for assisting the growth of crops during cold weather hydroponic cultivation. The present invention relates to a device which is installed on the outside of a pot and comprises: a structure in which a peltier element is placed on a mesh sheet of a cooper material, and an insulation is covered thereon; a control part which can control the temperature of a pad; a temperature sensor which detects the

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outside temperature.

CN110833292A. Water-circulating food cooler. The invention discloses a water-circulating food cooler. The food cooler at least comprises a cooling bowl and a water supply pipeline; the cooling bowl is of a hollow structure and divided into three layers of an inner layer, an interlayer and an outer layer; the interlayer partitions the hollow part of the cooling bowl into an inner cavity and an outer cavity, and two cavities are in interpenetration at the top of the interlayer; a hole is formed in the bottom of the interlayer, and the hole opening is a water inlet, and an outer let at the bottom of the outer cavity of the hollow part of the cooling bowl is a water outlet, the water inlet, the inner cavity, the outer cavity and the water outlet of the cooling bowl form a water-circulating channel of the cooling bowl; the water supply pipeline is arranged below the bottom of the cooling bowl and provided with a water feeding pipeline and a water returning pipeline, the water feeding pipeline of the water supply pipeline is butted joint with the water inlet of the cooling bowl, and the water returning pipeline is butted joint with the water outlet of the cooling bowl. Through the water-circulating food cooler disclosed by the invention, a dedicated cooling bowl is butted joint with the water supply pipeline, the circulating water enters the inner cavity of the cooling bowl to quickly take away the heat of the food, so that the cooler is safe and sanitary, convenient and comfortable to use, and suitable for use in large volumes.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

The conceptualization of the study is based on the concept of evaporative cooling methods specifically on the evaporative cooler made with clay and other porous materials. The clay pot cooling system will serve as cooling technology in the management of nutrient solution for indoor hydroponics production with the concept of thermodynamics and engineering. Cooling system has the vital role in hydroponic cultivation as may affect the growth and yield of the plants. Cooling the ambient air temperature of the whole greenhouse is quite expensive. Thus, cooling the nutrient solution up to the optimum condition is more effective.

Ndukwu (2011) stated that that the evaporative cooler can reduce the daily maximum ambient temperature from 32 - 40 °C to 24 – 29 °C i.e., a temperature reduction of up to 10 °C and increase the relative humidity of incoming air from 40.3% to 92% for the storage chamber.

The study will be using clay pot filled with water in which the tiny pores of the said clay pot serve as passage of small amount of water (taking away the high-velocity tail of the kinetic energy distribution) that creates cooling of water inside the pot (evaporative cooling) wherein the rolled pipe is submerged to said water inside the pot then the nutrient solution that passes through the rolled pipe is cooled through conduction and convection.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.

Designing a Clay Pot Cooling System

The preliminary design of the clay pot cooling system (CPCS) will use the basic heat transfer and fluid flow equations to be able to calculate the convective and conductive heat transfer coefficient and the length of the pipe. The water temperature inside the clay pot will be measured. The inlet temperature of the nutrient solution will depend on the surrounding condition.

To simplify the computations of the cooling system, thermo-physical properties of materials are assumed to be constant and steady-state-process. The parameters to be considered in system design will be:

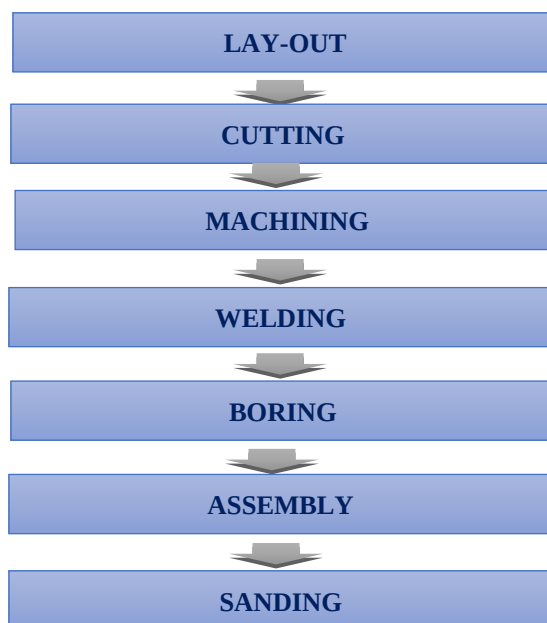
1. Length of pipe, pipe size, pipe types and arrangement, and water velocity;
2. Climatic condition (temperature inside the greenhouse and the water temperature inside the clay pot);
3. Thermo-physical properties of water, and pipe; and
4. Convective and conductive heat transfer resistance.

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Fabrication of an Clay Pot Cooling System

The materials needed in the study will be purchased from Aklan and Capiz through the fund of the Research Department. The earthen pot will be made to order in Leso, Aklan according to the design of the researcher and will undergo in processes: molding, kiln firing and drying. The frame and other parts of the cooling system will be fabricated by the researcher at Aranguel, President Roxas, Capiz with the aid of his brothers (welder/fabricator).

A sketch of the design will be provided. The fabrication will go through the following processes: lay-outting, cutting, welding, boring, riveting, bending, assembly and sanding.



The retrofitting of the room for indoor located in the Agricultural and Biosystems Engineering Department will also be considered.

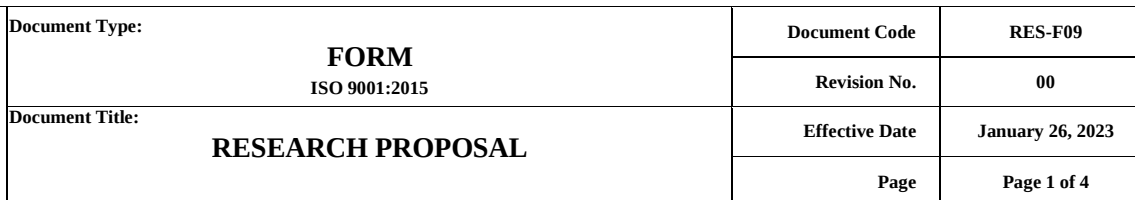
Testing and Evaluation of a Clay Pot Cooling System

Description and Operation of the System

The main components of CPCS are nutrient solution tank, earthen pot, submerged pipes, frame, cover, automatic water level controller, float ball valve and water pump. Each treatment will utilize its own submersible water pump to prevent mixing the nutrient solution. The 16-watt water pump will run continuously from 6:00 AM to 6:00 PM and will be accompanied by an automatic shut on/off mechanical timer to prevent overheating with an interval of 15 minutes or depending on the growth of cucumber. When the water pump is turned on, the nutrient solution from the tank will be pump out and passes through the submerged pipes in which cooling takes place then distributed to the growing bed. The overflowing nutrient solution from the growing bed which is affected by the ambient air temperature of the greenhouse will return back to the nutrient solution tank.

System Test-Run

The first trial testing of the system will be the arrangement of the set-up or system components. Another trial test will be conducted using the different types of pipe, rolled and submerged to water inside the clay pot to determine the most suitable material in terms of quality, flexibility and capacity to lower the nutrient solution temperature. For final set-up, pre-testing of the system will be performed using the recommended type of pipe with an empty clay pot to identify the possible leak in pipe connections. 10 days temperature reading will be collected to be able to test the effectiveness of the system.



Important parameters for growing lettuce, as well as environmental factors will be gathered, collected and recorded as the experiment progresses to properly acquire the information required in growing in indoor hydroponics. The parameters are the following:

- Yield parameters of lettuce such as fruit length (cm) and fruit fresh weight (g) will be measure after harvesting.

The yield was determined for each plant and the average was computed for each treatment. The average yield per treatment will be computed using the formula:

where: Y_{average} = Average yield per plant, (gram/plant)
 FW_{total} = Total fresh weight in one treatment, (g)
 NP_{total} = Total number of sample plants in one treatment

The data collected will be analyzed using the Statistical tool. The effectiveness between the CPCS and Control will be analyzed using t-test of two-sample assuming equal variances.

The yield parameters as influenced by different nutrient solution management methods will also be analyzed using t-test.

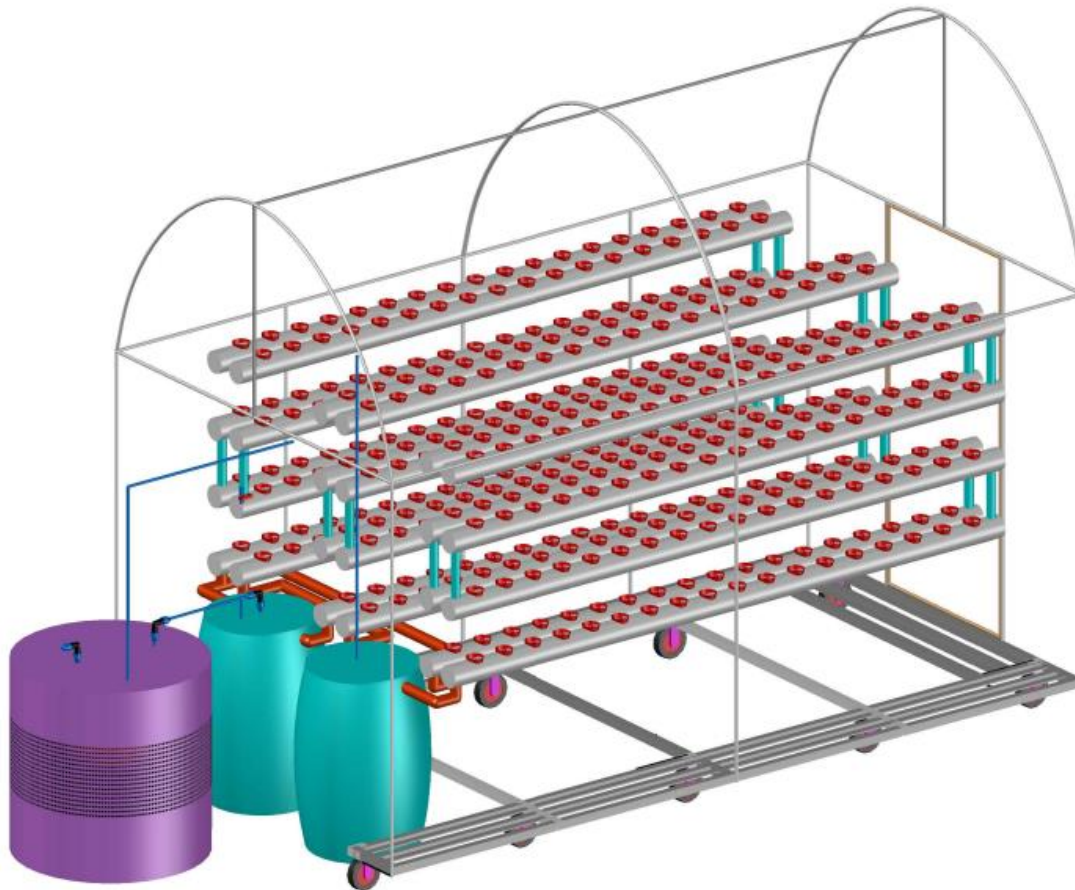
The total cost of production of using the system will be determined by calculating the fixed costs which includes the cost of the greenhouse and hydroponic system; and variable costs which includes the repair and maintenance, seeds, labor, electricity, nutrient solution and other miscellaneous cost. Net income, ROI and payback period will be included in the computation.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use Gantt chart.

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(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)



Clay Pot Cooling System for Indoor Hydroponics

(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

(17) Target Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

The target beneficiaries of the said technology are the Filipinos involve in hydroponics, aeroponics and aquaponics. This will also support to the program of the Department of Agriculture (DA) and other agencies in urban farming and an initiative to address soil degradation which resulted to unutilized agricultural land. In fact, agricultural land area in Western Visayas is decreasing and one of the reasons is continued agricultural lands conversion due to urban development or expansion. Also, soil is becoming crucial due to chemical reaction and degradation. According to PSA (2018), the farm area in Western Visayas declined from 754.4 thousand hectares in 1991 to 666.9 thousand hectares in 2002. In 2012, it has been recorded 467.8 thousand hectares and the trend is declining. Lastly, this technology will serve as basis for inventors, researchers especially to the students in the improvement of technology.

(18) Personnel Requirement

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(19) Budgetary Requirement

PROPOSED BUDGET

Construction Materials (Cooling System)	P 10,098.00
Hydroponics	83,706.00
Testing Materials (Nutrient Solution, lettuce, etc.)	6,000.00

TOTAL	P 99,855.00

(20) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Al-Rawahy, M.S., Al-Rawahy, S.A., Al-Mulla, Y.A., Nadaf, S.K. (2018). Effect of Cooling Root-Zone Temperature on Growth, Yield and Nutrient Uptake in Cucumber Grown in Hydroponic System During Summer Season in Cooled Greenhouse. *Journal of Agricultural Science*. 11(1), 47-60. <https://doi.org/10.5539/jas.v11n1p47>

Chukwuneke, J.L., Ajike, C.O., Achebe, C.H., Okolie, P.C., (2012). A Mathematical Model of an Evaporative Cooling Pad Using Sintered Nigerian Clay. *Journal of Minerals and Materials Characterization and Engineering*, 11, 1113-1120. Retrieved from <https://www.SciRP.org/journal/jmmce>

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Munar, R.M., (2019). In-Ground Passive Cooling System: Its Application in the Management of Nutrient Solution for Hydroponic Production of Basil (*Ocimum basilicum* L.) (Master's thesis). Central Luzon State University, Science City of Muñoz, Nueva Ecija, Philippines.

Ndukwu. M. C. (2011) Development of a Low Cost Mud Evaporative Cooler for Preservation of Fruits and Vegetables. *Agricultural Engineering International: CIGR Journal*. Manuscript No.1781. 13(1), 1-8. Retrieved from www.academicjournals.com

Nicolov, N.V., Atanasov, A.Z., Evstatiev, B.I., Vladut, V.N., Biris, S.S., (2023). Design of a Small-Scale Hydroponic System for Indoor Farming of Leafy Vegetables. *Agriculture* 2023, 13, 1191. <https://doi.org/10.3390/agriculture13061191>

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Shafie, A.M.M., el al., (2022). Effect of Various Photoperiods Towards Lollo Bionda Growth using IoT Based Lighting Control for Indoor Hydroponic Farming System. *IOP Conf. Ser.: Earth Environ. Sci.* 1114 012069

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Appendix Table 1. Estimated Budgetary Requirement of Cooling System

Quantity	Unit	Description	Unit Price (P)	Amount (P)
1	pieces	clay pot (30 in x 30 in)	3000.00	3000.00
1	pieces	Stainless Steel Plate	1500.00	1500.00
1	pieces	Straight PVC pipe 1/2"	50.00	50.00
4	pieces	elbow adapter black 1/2"	90.00	360.00
80	meter	rolled PVC pipe 1/2"	20.00	1600.00
2	pieces	PVC ball valve 1/2"	100.00	200.00
1	pieces	water pump (16 watt)	1500.00	1500.00
1	can	pipe cement (1/2 liter)	500.00	500.00
1	load	GI pipe 1/2"	1400.00	1400.00
1	kg	welding rod	120.00	120.00
1	can	vulcaseal	500.00	500.00
3	pieces	nylon cable	95.00	285.00
Material Cost (EPCS)				6,515.00
Labor				2,931.75
Contingencies (10% of Total Material Cost)				651.50
Total				10,098.00

Appendix Table 2. Estimated Budgetary Requirement of Testing Materials

Quantity	Unit	Description	Unit Price (P)	Amount (P)
5	pack	Lollo Biond lettuce seeds	130.00	650.00
12	pieces	nutrient solution	450.00	5400.00
Testing Material Cost (EPCS and Control)				6,050.00

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Appendix Table 3. Estimated Budgetary Requirement for Hydroponics

Quantity	Unit	Description	Unit Price (P)	Amount (P)
2	pieces	polyethylene plastic drum (30 gallon)	1100.00	2200.00
10	pieces	Straight PVC pipe 1/2"	50.00	500.00
6	pieces	tee 1/2"	15.00	90.00
18	pieces	elbow 1/2"	10.00	180.00
6	pieces	end cap 1/2"	10.00	60.00
4	pieces	elbow adapter black 1/2"	90.00	360.00
2	pieces	PVC orange 2"	155.00	310.00
6	meter	elbow 2"	65.00	390.00
2	pieces	tee 2"	65.00	130.00
2	pieces	cleanout 2"	65.00	130.00
32	pieces	PVC pipe orange 4"	500.00	16000.00
64	pieces	PVC end cap 4"	80.00	5120.00
2	pieces	PVC ball valve 1/2"	100.00	200.00
2	pieces	water pump (54 watt)	3000.00	6000.00
1	can	pipe cement (1/2 liter)	500.00	500.00
1	load	GI pipe 1/2"	1400.00	1400.00
1	load	GI pipe 3/4"	1400.00	1400.00
1	kg	welding rod	120.00	120.00
500	pieces	net pots	10.00	5000.00
1	can	vulcaseal	500.00	500.00
4	pieces	nylon cable	100.00	400.00
1	roll	nylon wire	300.00	300.00
25	meter	shade net	100.00	2500.00
32	units	LED lights	500.00	16000.00

Material Cost (Retrofitting of greenhouse and hydroponics)	59,790.00
Labor	17,937.00
Contingencies (10% of Total Material Cost)	5,979.00
Total	83,706.00